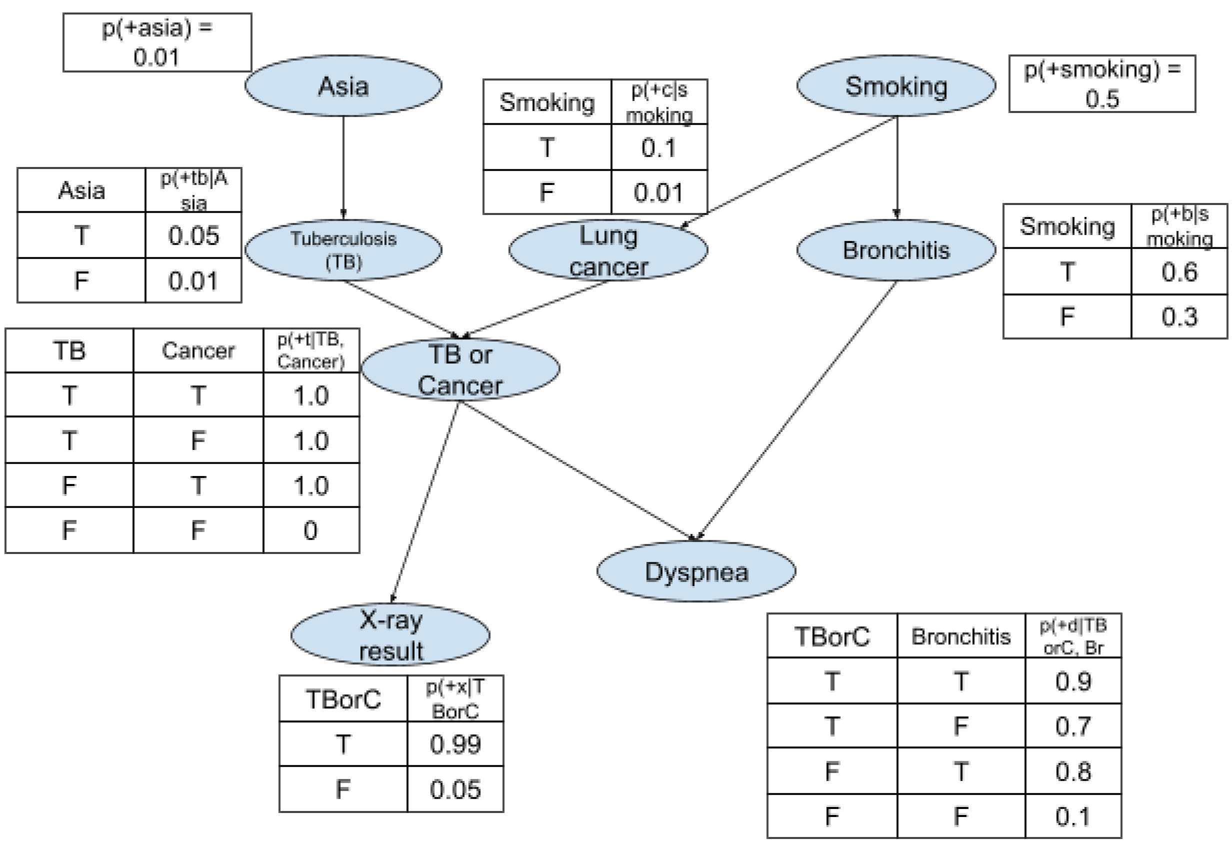
Programming project 1: Bayesian networks

Due: see Canvas
Group work: You may work in groups of 1-3. Include all group members when submitting to Gradescope.

The goal of this project is to complete a Python program that can be used to diagnose between three lung diseases - Tuberculosis, Lung cancer, and Bronchitis - based on any combination of symptoms. The diagnosis is to be made using the same Bayesian network shown during the lecture[1]:



The GUI portion of the program has been completed. You are to write a Python class, `Diagnostics`, that **must** use the textbook's code to provide the Bayesian network logic. Your class `Diagnostics` should have a constructor (that takes no parameters) and a single method:

```
diagnose (visit_to_asia, smoking, xray_result, dyspnea)
```

[1] From D.M. Buede, J.A. Tatman, T.A. Bresnick "Introduction to Bayesian Networks," Tutorial for the 66th MORS Symposium

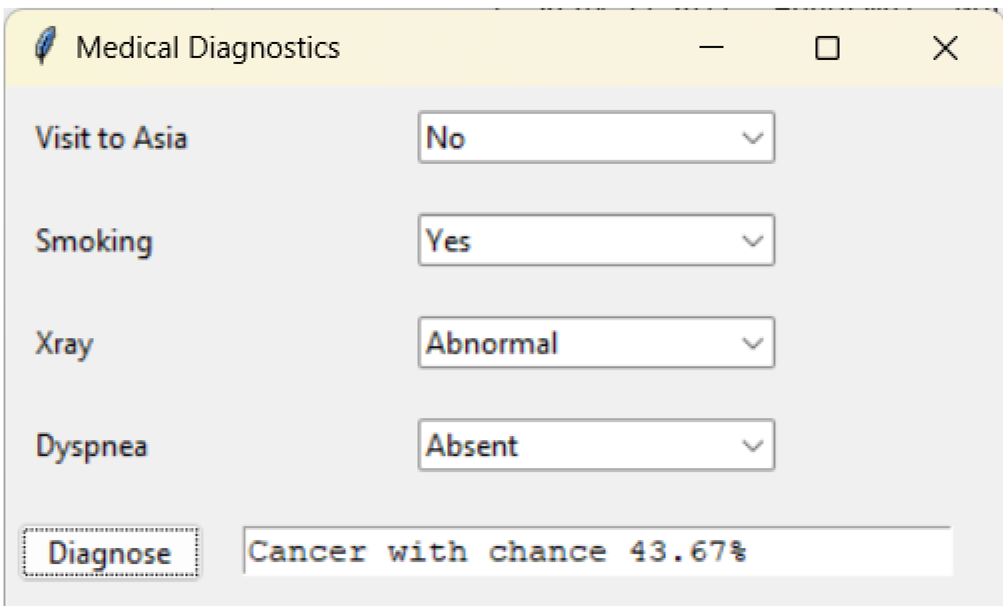
`visit_to_asia, smoking, xray_result, dyspnea` are strings that describe the symptoms. If a symptom is not available, it has the value "NA". The possible values are:

1. `visit_to_asia: Yes, No, NA`
2. `smoking: Yes, No, NA`
3. `Xray_result: Abnormal, Normal, NA`
4. `Dyspnea: Present, Absent, NA`

The method should return a list of two items:

1. The name of the disease that is most likely given the symptoms (a string): "TB", "Cancer", or "Bronchitis". You must use exactly these words.
2. The probability of the most likely disease (a number between 0 and 1).

The Diagnostics class will be called by the GUI where you can set the symptoms and see the diagnosis.



Hints:

- The completed GUI code and a skeleton of the diagnostics class are on Canvas: `diagnostics_gui.py` and `diagnostics.py`.
- You will need to download probability4e.py, utils4e.py from https://github.com/aimacode/aima-python You will also need to install the numpy package.
- Your class should make use of [class BayesNet](#) in the textbook code. You will create your BayesNet in the constructor. You should not use any other Bayesian network library.
- Use the enumeration inference algorithm to calculate the probability of each disease. This algorithm is implemented as the [enumeration_ask function](#) in the textbook code.

- The textbook class BayesNet only handles Boolean variables. So, you will have to write code to convert Yes/No, Abnormal/Normal, Present/Absent to True/False.
- You are allowed (and encouraged) to use AI tools (such as ChatGPT, Copilot) to write your code. However, you are still responsible for understanding the code and the quality of the code (i.e., overly complicated code can be graded lower).

Submit via Gradescope:

Submission is via Gradescope. Upload a single .py file with your class. The name of the file **must be** `diagnostics.py`. The name of the class **must be** `Diagnostics`. These are case-sensitive. Gradescope will run a few unit tests automatically and show you the results. Include all group members when submitting to Gradescope.